Environment-dependent peer pressure in spatial public goods games with reinforcement learning

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Peer pressure is a common form of social influence and affects the evolution of cooperation in social dilemmas. Existing models usually treat peer pressure as a rule that does not change with the local behavioral environment, whereas in reality the behavior that receives pressure depends on what most members of a group do. How such pressure interacts with individual learning remains unclear. To address this, we propose an environment-dependent peer pressure mechanism in a spatial public goods game with reinforcement learning. Defectors receive pressure in groups with enough cooperators, and cooperators receive pressure in the other groups. Individuals update their actions by Q-learning, and their state includes both their own action and the environment of the current group. We find that weak pressure allows cooperation to emerge at lower synergy factors, whereas strong pressure yields higher cooperation levels at higher synergy factors. For a given synergy factor, cooperation increases with the pressure intensity up to an optimal value and then drops abruptly to zero. Under moderate pressure, cooperative clusters survive and individuals learn to cooperate in cooperative environments. Under strong pressure, such environments disappear before this preference can be learned. In addition, the threshold that defines a cooperative environment changes the response of cooperation to pressure, and a state that includes both the action and the environment supports cooperation over a wider range of synergy factors than a state with only one of them. This study shows how local social feedback and individual learning jointly affect the evolution of cooperation.

Social disapproval can change how people behave in groups. However, the behavior that is disapproved of is not always the same. In a group where most members contribute to a common good, those who do not contribute are criticized. In a group where few members contribute, the contributors are the ones who differ from the others. We include this feature in a public goods game on a lattice, where individuals learn from their own experience through reinforcement learning and can tell whether their current group is mostly cooperative. We find that moderate pressure helps clusters of cooperators to survive and expand, so that cooperation finally reaches a high level. When the pressure is too strong, cooperative groups disappear at an early stage and cooperation does not recover. The outcome also depends on how many cooperators are required for a group to be regarded as cooperative and on the information that individuals use for learning.

I. INTRODUCTION

Cooperation is common in nature and human society, and it is essential for the functioning of groups ranging from microbial communities to human organizations [1, 2]. However, cooperative behaviors require individuals to bear a cost, while the resulting collective benefits can be exploited by others [3]. Understanding how cooperation can emerge and persist despite the advantage of selfish behaviors has therefore become a central topic in many disciplines [4].

Evolutionary game theory is the standard framework for studying this problem [5–7]. For interactions in groups, the public goods game (PGG) is the most commonly used model [8–10]. In spatial PGGs, individuals interact only with their neighbors, so that cooperators can form clusters and resist exploitation [11–17]. Spatial structure alone, however, sustains cooperation only when the return of the public good is high enough. Additional mechanisms have therefore been introduced [18], such as punishment [19–23], reward [24, 25], reputation [26, 27], and environmental feedback [28–31]. These mechanisms change the payoffs of cooperators and defectors, and thereby the conditions under which cooperation can spread.

Besides such incentives, the attitudes of other group members also affect individual behavior. Peer pressure is the influence that a group exerts on its members to behave like the others. Individuals whose behavior differs from that of the people around them may face, or expect to face, disapproval or exclusion, and they tend to adjust their behavior accordingly [32, 33]. In evolutionary games, peer pressure is usually modeled as a reduction of payoff, in two ways. In the first, two neighbors with different strategies penalize each other, so that each individual loses payoff in proportion to the number of its neighbors with the opposite strategy, whether it cooperates or defects [34]. This rule has been extended to interdependent networks [35] and to individuals with different or time-varying sensitivities to pressure [36, 37]. In the second, cooperation is regarded as the social norm, and only defectors lose payoff, in proportion to the number of cooperators in their group [38]. Both rules can promote cooperation [34, 36, 38]. In both rules, however, the behavior that receives pressure is specified in advance and is the same in every group. Under the first rule, both behaviors are penalized wherever they meet, and only the amount of the penalty depends on the neighbors. Under the second rule, cooperators are never penalized, whatever the composition of their group.

In reality, however, the same behavior can receive differ-

ent social feedback in different groups. Whether a behavior is approved of depends on how widespread it is in the group concerned. People tend to follow the behavior that is common around them [33, 39–41], and a behavior is judged as more acceptable, and as less deserving of punishment, when it is common than when it is rare, whether it is altruistic or selfish [42]. In a team whose members generally work hard, a member who contributes little is criticized. The opposite case has been observed in public goods experiments. Participants punish not only free riders but also those who contribute more than the others, most often where norms of cooperation are weak [43]. This punishment becomes stronger when the contributions of the other members are more alike [44], and members who give much more than the rest are disliked because they are seen as setting an undesirable standard [45]. The level of cooperation in a group thus acts as a local environment that determines which behavior receives pressure. This environment is not fixed. It changes with the actions of the group members, as in games with environmental feedback [28–31], except that here each group has its own environment. Moreover, the level of cooperation at which the expectations of a group change need not be one half. Experiments on social conventions have shown that a group can adopt a new convention once a committed minority reaches a critical size [46].

Studying such pressure also requires a description of how individuals respond to their environment. Traditional evolutionary game models assume that individuals imitate more successful neighbors [13, 47]. In contrast, reinforcement learning describes individuals who adjust their behavior according to the rewards of their own past actions [48, 49], and it has been increasingly used in evolutionary games [50, 51]. In particular, Q-learning [52] has been applied to PGGs and other social dilemmas [53–57], also in combination with punishment, reward, and reputation [58–62]. In most of these studies, the state of an individual is its own previous action [63–65]. Recent studies have shown that information about the environment also affects cooperation [66–68]. When peer pressure depends on the group environment, it is natural to include this environment in the state.

In this work, we propose an environment-dependent peer pressure mechanism in a spatial PGG with Q-learning. A group is a cooperative environment if its number of cooperators reaches a threshold, and a defective environment otherwise. Defectors receive pressure in cooperative environments, and cooperators receive pressure in defective environments. The state of each individual consists of its own action and the environment of its current group, so that it can learn different actions for different environments. We find that weak pressure allows cooperation to emerge at lower synergy factors, whereas strong pressure yields higher cooperation levels at higher synergy factors, and that cooperation collapses once the pressure exceeds an optimal intensity. Spatial patterns and learning dynamics show that this collapse occurs because strong pressure removes cooperative environments before individuals learn to cooperate in them. The threshold further changes the response of cooperation to pressure.

II. MODEL

A. Spatial public goods game

We consider a spatial public goods game on a square lattice with periodic boundary conditions. The population consists of $L \times L$ individuals, with each individual occupying a single lattice site. Each interaction group consists of a focal individual and its four nearest neighbors, so that the group size is $g = 5$. Each individual therefore belongs to five overlapping groups, including the group centered on itself and the groups centered on its four neighbors.

Each individual i adopts one of two actions, $a_i \in \mathscr{A} = \{\mathrm{C}, \mathrm{D}\}$, where C denotes cooperation and D denotes defection. In each group interaction, cooperators contribute a fixed cost $c = 1$, whereas defectors contribute nothing. The total contribution is multiplied by the synergy factor r and equally distributed among all group members. The payoff of individual i in a group interaction is

$$\pi_i = \frac{r n_{\mathrm{C}}}{g} - c_i, \tag{1}$$

where n_{C} is the number of cooperators in the group, and $c_i = 1$ if individual i cooperates and $c_i = 0$ otherwise.

B. Environment-dependent peer pressure

We assume that peer pressure depends on the local environment of each interaction group. The environment is determined separately for each group according to the environmental threshold T, which is an integer between 0 and g. When the number of cooperators in the group satisfies $n_{\mathrm{C}} \geq T$, the group is in a cooperative environment, denoted by G. Otherwise, it is in a defective environment, denoted by B.

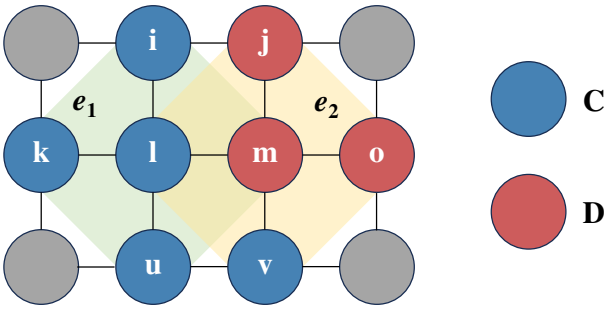
The environment determines which behavior receives pressure. In a cooperative environment, defectors receive pressure from the cooperators in the group. In a defective environment, cooperators receive pressure from the defectors in the group. The latter case represents the disapproval of behavior that differs from what is typical in a group, as observed in public goods experiments [43–45]. It does not imply that defection is desirable for the group. The peer pressure on individual i is defined as

$$P_i = \begin{cases} -\beta\, n_{\mathrm{C}}, & a_i = \mathrm{D},\ n_{\mathrm{C}} \geq T, \\ -\beta\, n_{\mathrm{D}}, & a_i = \mathrm{C},\ n_{\mathrm{C}} < T, \\ 0, & \text{otherwise}, \end{cases} \tag{2}$$

where $n_{\mathrm{D}} = g - n_{\mathrm{C}}$ is the number of defectors in the group and β is the intensity of peer pressure. Here, a_i and n_{C} refer to the actions implemented in the current interaction. The effective reward of individual i in a group interaction is

$$R_i = \pi_i + P_i. \tag{3}$$

Figure 1 illustrates these definitions for two overlapping groups and $T = 3$, which for $g = 5$ corresponds to the simple-majority rule. Two limiting cases are worth noting. For $T \leq 1$,



e_1	π	P	s
i	0.8r-1	0	s_1
k	0.8r-1	0	s_1
l	0.8r-1	0	s_1
m	0.8r	-4β	s_3
u	0.8r-1	0	s_1

$n_C \geq 3, e_1 = \mathrm{G}$

e_2	π	P	s
j	0.4r	0	s_4
l	0.4r-1	-3β	s_2
m	0.4r	0	s_4
o	0.4r	0	s_4
v	0.4r-1	-3β	s_2

$n_C < 3, e_2 = \mathrm{B}$

FIG. 1. The environment of a group determines the payoff, peer pressure, and state of its members. Two overlapping interaction groups are shown for $T = 3$. Blue and red circles denote cooperators and defectors, and gray circles are individuals outside the two groups. Group e_1, centered on l, contains four cooperators and is a cooperative environment (G), so its defector m receives pressure. Group e_2, centered on m, contains two cooperators and is a defective environment (B), so its cooperators l and v receive pressure. The tables list the payoff π, the peer pressure P, and the state s of each group member. Individuals l and m belong to both groups and have different states in them.

a group can be defective only if it contains no cooperator, so cooperators are never pressured. This corresponds to the assumption that cooperation is the norm in every group [38]. The cases $T = 0$ and $T = 1$ then differ only in whether a fully defecting group is distinguished as a separate environment. For $T = g$, a group is cooperative only if it contains no defector, so defectors are never pressured.

C. Q-learning with environmental information

To describe how individuals adapt their behavior, we introduce Q-learning [49, 52] into the spatial public goods game. Previous studies have often used the previous action of an individual to define its state [63–65]. However, including environmental information in the state allows individuals to distinguish group environments during learning [68]. In our model, the effective reward depends on both the action of an individual and the environment of the group. The state is therefore defined by the current action of the individual and the environment of the interaction group. Combining these two components gives four states, $s_1 = (\mathrm{C},\mathrm{G})$, $s_2 = (\mathrm{C},\mathrm{B})$, $s_3 = (\mathrm{D},\mathrm{G})$, and $s_4 = (\mathrm{D},\mathrm{B})$.

For each individual, the Q-table records the expected discounted cumulative reward of each state-action pair and has the form

$$Q_i = \begin{bmatrix} Q_i(s_1,\mathrm{C}) & Q_i(s_1,\mathrm{D}) \\ Q_i(s_2,\mathrm{C}) & Q_i(s_2,\mathrm{D}) \\ Q_i(s_3,\mathrm{C}) & Q_i(s_3,\mathrm{D}) \\ Q_i(s_4,\mathrm{C}) & Q_i(s_4,\mathrm{D}) \end{bmatrix}, \tag{4}$$

where the two columns correspond to the two actions. Because the environment is defined for each group, the same individual is generally in different states in the different groups to which it belongs (see individuals l and m in Fig. 1), but it uses a single Q-table in all of them.

In state s, an individual selects its action according to the ε-greedy rule. With probability $1-\varepsilon$ it chooses the action with the higher Q-value in the current state, and with probability ε it chooses one of the two actions at random. When the two Q-values are equal, one action is selected at random. After all members of the group have implemented their actions, individual i receives the effective reward R_i and updates its Q-table according to

$$Q_i(s,a) \leftarrow (1-\alpha)\,Q_i(s,a) + \alpha\left[R_i + \gamma \max_{a' \in \mathscr{A}} Q_i(s',a')\right], \tag{5}$$

where α is the learning rate and γ is the discount factor. The new state s' is determined by the action just implemented and the environment of the same group after all its members have acted.

D. Simulation procedure

Initially, each individual is assigned C or D with equal probability, and all Q-values are set to zero. We use a single group interaction as the elementary unit of learning, rather than accumulating payoffs over all overlapping groups [68]. In each elementary step, a focal individual is randomly selected, and the group centered on it plays the game once. All members of this group select their actions, receive their effective rewards, and update their Q-tables according to the environment of this group. One Monte Carlo step (MCS) consists of L^2 elementary steps, so that each individual participates in g group interactions on average. The complete procedure is summarized in Algorithm 1.

Algorithm 1 Spatial public goods game with peer pressure and Q-learning.

Input: L, r, T, β, α, γ, ε, and the number of Monte Carlo steps $t_{\max}$

for each individual i **do**
  Set a_i to C or D with equal probability;
  Set $Q_i(s,a) = 0$ for all s and a;
end for
for $t = 1$ to $t_{\max}$ **do**
  for $n = 1$ to L^2 **do**
    Randomly select a focal individual; let Ω be the group formed by it and its four nearest neighbors;
    Determine the environment e of Ω;
    for each individual $j \in \Omega$ **do**
      $s_j \leftarrow (a_j, e)$;
      With probability ε, select a'_j randomly from $\mathscr{A}$; otherwise select the action a'_j with the higher Q-value in state s_j (randomly if the two values are equal);
    end for
    Implement all actions simultaneously, $a_j \leftarrow a'_j$ for all $j \in \Omega$;
    Determine the new environment e' of Ω;
    for each individual $j \in \Omega$ **do**
      Calculate π_j, P_j, and R_j by Eqs. (1)–(3);
      $s'_j \leftarrow (a_j, e')$;
      Update $Q_j(s_j, a_j)$ by Eq. (5);
    end for
  end for
end for

The cooperation level ρ_C is the fraction of cooperators in the population. Its stationary value is calculated by averaging over the last 10^5 MCS of a simulation lasting 10^6 MCS. For parameter sets that do not reach a stationary state within 10^6 MCS, the simulation is extended to 10^8 MCS, and the average is taken over the last 10^7 MCS. Each data point is averaged over 20 independent runs. Unless stated otherwise, we use $L = 100$, $T = 3$, and the learning parameters $\alpha = 0.1$, $\gamma = 0.9$, and $\varepsilon = 0.01$ [53, 58, 68].

III. RESULTS AND DISCUSSION

A. Interplay of pressure intensity and synergy factor

We first investigate how peer pressure affects cooperation under different synergy factors. Figure 2(a) shows the cooperation level ρ_C as a function of r for different values of β. Without pressure, cooperation emerges at the lowest synergy factor and increases gradually with r. As β increases, the onset of cooperation shifts toward higher r. When r is small, a smaller β therefore makes cooperation more likely to emerge. Once cooperation has emerged, however, a larger β produces a higher cooperation level, and the order of the curves at large r is the reverse of the order of their onsets. Weak pressure and strong pressure are thus each more favorable in a different range of r. In addition, the emergence of cooperation becomes sharper as β increases. The cooperation level rises gradually from zero for $\beta = 0$, whereas for larger β it jumps from zero to a high value within a narrow range of r.

Figure 2(b) shows ρ_C as a function of β for different values of r. For each r, the cooperation level first increases with β. As β increases, the curves for different r approach each other, which indicates that the cooperation level is mainly determined by β as long as cooperation is maintained. For $r < 4$, there is an optimal β that maximizes cooperation. Beyond this value, cooperation collapses abruptly to zero, in a manner similar to a discontinuous phase transition [16]. Both the optimal β and the collapse shift toward larger β as r increases. For $r = 4.0$, cooperation is maintained up to the largest β examined.

These observations can be understood from the difference between the effective rewards of cooperation and defection in a single interaction, which is derived in the Appendix. According to Eq. (A.1), peer pressure raises the reward of cooperation relative to defection in groups where most co-players cooperate, and lowers it in groups where most co-players defect. Pressure therefore strengthens the behavior of the local majority. When r is small, the cost of cooperation is high and defectors more easily become the local majority, so that stronger pressure suppresses cooperation. When r is large, cooperative groups are easier to maintain, and stronger pressure on the defectors in these groups raises the cooperation level. The same reasoning explains why a larger r allows cooperation to persist under stronger pressure.

B. Time evolution and spatial patterns

Figure 3 presents the time evolution of cooperation and the corresponding spatial patterns at $r = 3.5$, comparing the cases without pressure ($\beta = 0$), with moderate pressure ($\beta = 0.1$), and with strong pressure ($\beta = 0.2$).

As shown in Fig. 3(a), cooperation initially decreases in all three cases, and stronger pressure produces a faster decline. Without pressure, cooperation shows only a limited recovery after the decline. At $\beta = 0.1$, cooperation reaches a lower minimum than without pressure but subsequently recovers to a higher level. At $\beta = 0.2$, cooperation rapidly approaches zero and does not recover. Moderate pressure therefore lowers cooperation at the early stage but raises it in the stationary state, whereas strong pressure drives the population toward defection.

Figure 3(b) shows the spatial patterns that accompany these trajectories. At $\beta = 0.1$, small cooperative clusters survive the initial decline, then expand, and finally occupy most of the lattice. At 5×10^4 MCS, the cooperation levels at $\beta = 0$ and $\beta = 0.1$ are comparable. However, cooperators and defectors remain finely mixed without pressure, whereas individuals with the same action form large clusters at $\beta = 0.1$. Under strong pressure, the lattice is already dominated by defectors at 3×10^3 MCS, and only a few isolated cooperators appear later because of exploration. Moderate pressure thus favors the survival and expansion of cooperative clusters, which provides a spatial basis for the recovery of cooperation. Strong pressure removes cooperators before such clusters can form.

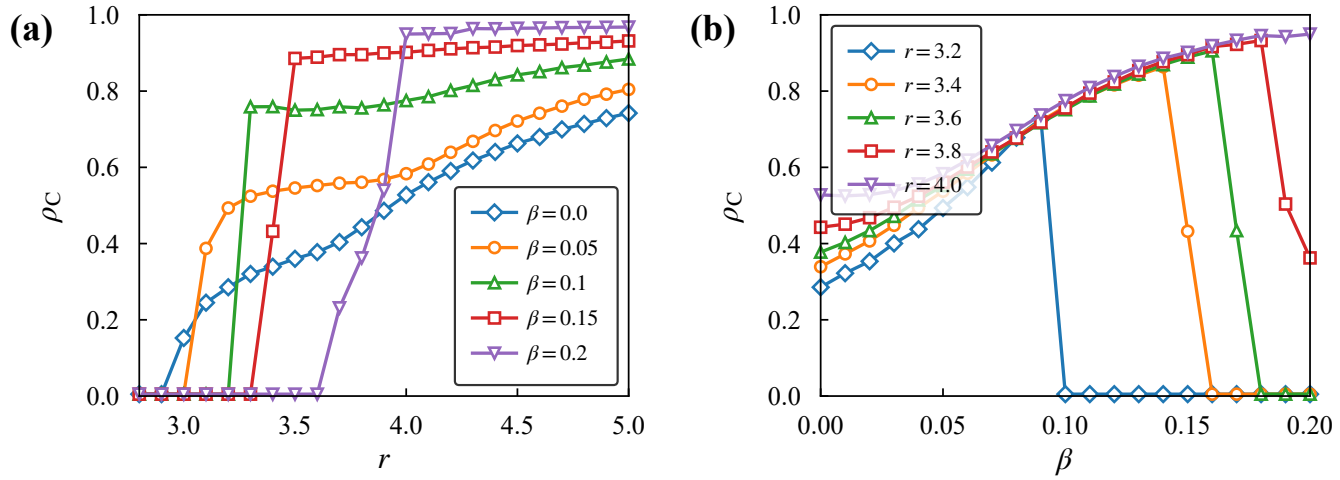


FIG. 2. Weak pressure favors the emergence of cooperation at small synergy factors, whereas strong pressure yields higher cooperation at large synergy factors. (a) Cooperation level ρ_C as a function of the synergy factor r for different pressure intensities β. (b) ρ_C as a function of β for different values of r. For $r < 4$, ρ_C reaches its maximum at an optimal β and then drops abruptly to zero.

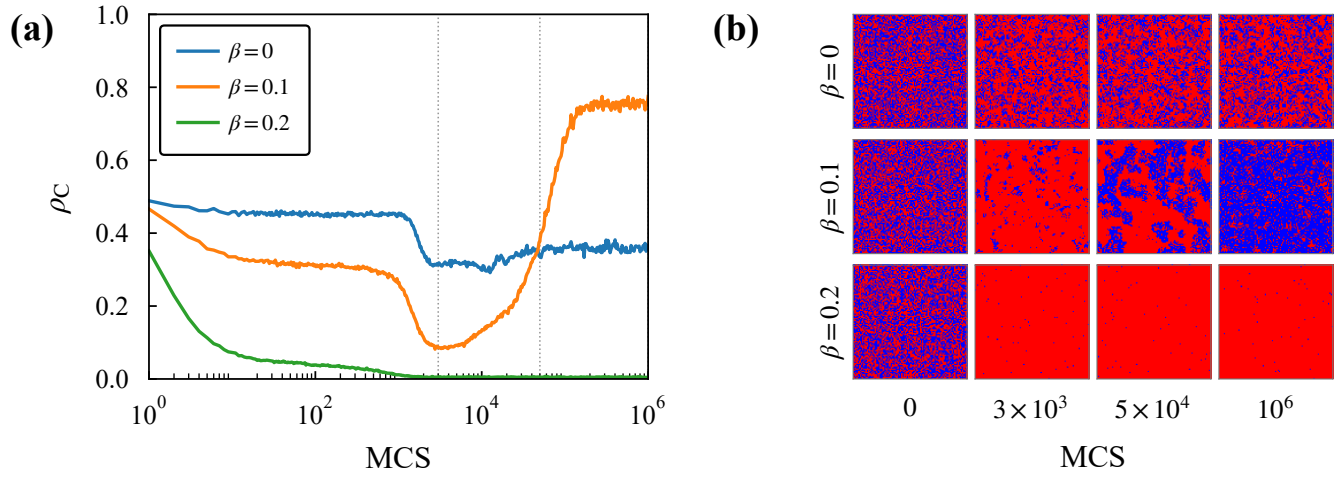


FIG. 3. Moderate pressure allows cooperative clusters to survive and expand, whereas strong pressure leads to defection. (a) Time evolution of the cooperation level ρ_C for $\beta = 0$, 0.1, and 0.2. The vertical dotted lines mark the times of the second and third snapshots. (b) Spatial snapshots at 0, 3×10^3, 5×10^4, and 10^6 MCS (from left to right) for $\beta = 0$, 0.1, and 0.2 (from top to bottom). Cooperators and defectors are shown in blue and red, respectively. For each β, the curve and the snapshots are obtained from the same simulation. The synergy factor is $r = 3.5$.

C. Learning dynamics under moderate and strong pressure

To further understand why moderate pressure promotes cooperation and strong pressure suppresses it, Fig. 4 examines how often each state is visited and which action is preferred in it. The preference is measured by the Q-value difference

$$A_s(t) = \left\langle Q_i(s,\mathrm{C}) - Q_i(s,\mathrm{D}) \right\rangle_{s,t}, \tag{6}$$

where $\langle\cdot\rangle_{s,t}$ denotes the average over all visits to state s during MCS t. Q-values are evaluated before the corresponding updates, and repeated visits by the same individual are counted separately. Positive and negative values of A_s indicate an average preference for cooperation and defection, respectively.

Under moderate pressure, Fig. 4(a) shows that A_{s_1} increases rapidly during the early stage, while A_{s_3} is first negative and later becomes positive. Cooperation thus becomes preferred, on average, in both states associated with cooperative environments. Although the overall cooperation level initially declines, s_1 and s_3 continue to be visited, because the surviving cooperative clusters shown in Fig. 3 keep providing cooperative environments. In these environments, defectors receive pressure while cooperators do not. These interactions allow individuals to gradually develop a preference for cooperation, including in s_3, where their current action is defection. In the stationary state, A_{s_2} is negative and A_{s_4} remains slightly negative, so that defection is preferred in defective environments.

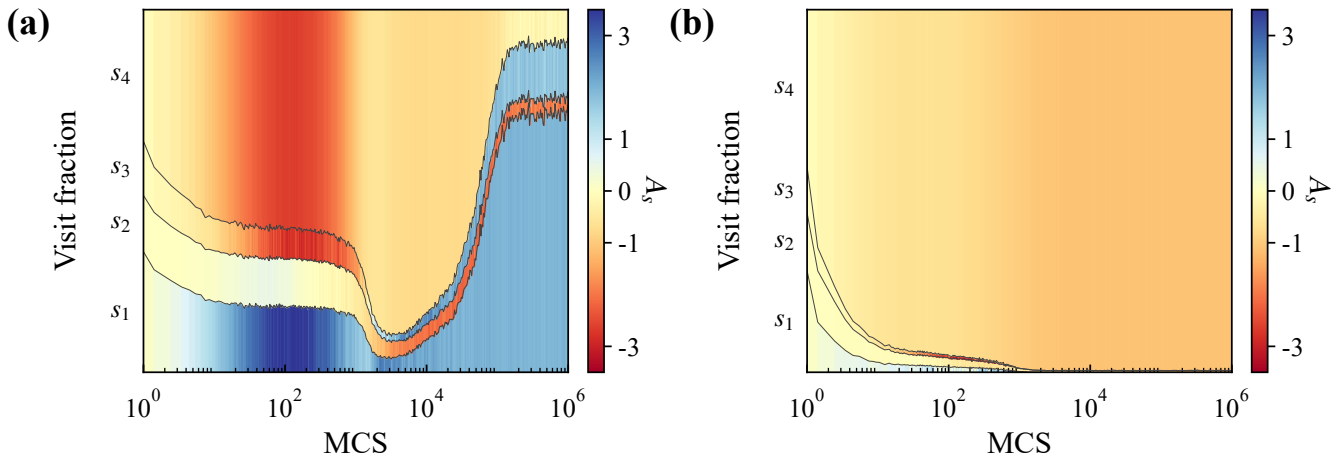


FIG. 4. Cooperative environments continue to be visited under moderate pressure but disappear under strong pressure. (a) $\beta = 0.1$. (b) $\beta = 0.2$. The four bands from bottom to top correspond to the states s_1, s_2, s_3, and s_4, and the width of each band gives the fraction of visits to that state. The color represents the Q-value difference A_s defined in Eq. (6), where blue and red indicate a preference for cooperation and defection, respectively. The color scale is nonlinear. The synergy factor is $r = 3.5$.

Individuals have thus learned to cooperate in cooperative environments and to defect in defective ones. The persistence of cooperative environments therefore retains the learning opportunities that support the recovery of cooperation.

Under strong pressure, Fig. 4(b) shows a rapid concentration of visits in s_4, which combines defection with a defective environment. With Q-values initially set to zero, the earliest updates are driven mainly by immediate rewards. Stronger pressure on cooperators in defective environments favors the early development of a preference for defection, which contributes to the faster decline of cooperation. Visits to s_1 and s_3 consequently become rare, leaving almost no opportunities for learning in cooperative environments. Meanwhile, A_{s_4} remains negative, indicating a persistent preference for defection. It should be noted that strong pressure does not reduce the reward of cooperation in cooperative environments, as shown by Eq. (A.1). Instead, it accelerates the loss of these environments, which limits the development of a preference for cooperation and prevents the recovery of cooperation.

D. Role of the environmental threshold

We next examine how the environmental threshold T affects cooperation at $r = 3.5$. Figure 5(a) shows that cooperation is highest at $T = 3$, followed by $T = 2$, under weak and moderate pressure. At $\beta = 0.2$, high cooperation occurs at $T = 0$, $T = 1$, and $T = 2$, whereas cooperation at $T = 3$ is close to zero. At $T = 4$ and $T = 5$, cooperation remains close to zero for all pressure intensities shown. Thus, intermediate thresholds favor cooperation under weak and moderate pressure, whereas lower thresholds are more favorable under strong pressure. The cooperation level also differs among thresholds at $\beta = 0$. Since the rewards do not depend on T in this case, this difference comes from the state definition alone, because T determines how individuals distinguish the two environments during learning.

Figure 5(b) shows how cooperation responds to pressure at each threshold. At $T = 1$, cooperation remains close to zero over most of the range and rises abruptly only when β is close to the largest value examined. Cooperators are always exempt from pressure at this threshold, but defectors face only weak pressure when cooperators are scarce. Only sufficiently strong pressure can constrain defection and enable cooperation to emerge. At $T = 2$, cooperation increases monotonically with β. Only two cooperators are needed to form a cooperative environment, so a cooperator receives pressure only when it is the single cooperator in its group. Stronger pressure then mainly increases the cost of defection and favors cooperation. At $T = 3$, cooperation first increases with β and then collapses, as analyzed in Secs. III A–III C. At $T = 4$, cooperation remains close to zero. A cooperative environment requires at least four cooperators, so cooperators receive pressure in most mixed groups and cooperation is difficult to sustain. These results show that the threshold changes both the pressure needed to promote cooperation and the extent to which cooperation can withstand stronger pressure.

E. Comparison of pressure rules and state representations

To examine the roles of the pressure rule and of the information available for learning, we compare two pressure rules and three state representations. The action set and the learning parameters remain unchanged in these comparisons.

Figure 6(a) compares the environment-dependent rule of Eq. (2) with an environment-independent rule, under which individuals receive pressure from all group members with the opposite action regardless of the group environment,

$$P_i^{\mathrm{ind}} = \begin{cases} -\beta\, n_{\mathrm{D}}, & a_i = \mathrm{C}, \\ -\beta\, n_{\mathrm{C}}, & a_i = \mathrm{D}. \end{cases} \tag{7}$$

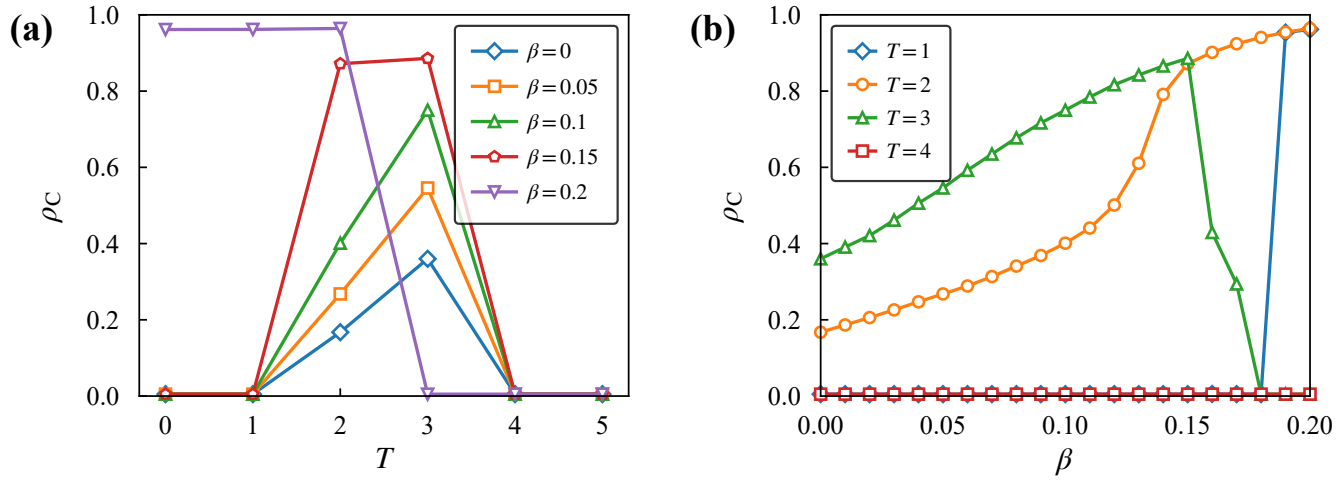


FIG. 5. The environmental threshold changes the response of cooperation to peer pressure. (a) Cooperation level ρ_C as a function of the environmental threshold T for different pressure intensities β. (b) ρ_C as a function of β for $T = 1$, 2, 3, and 4. The synergy factor is $r = 3.5$.

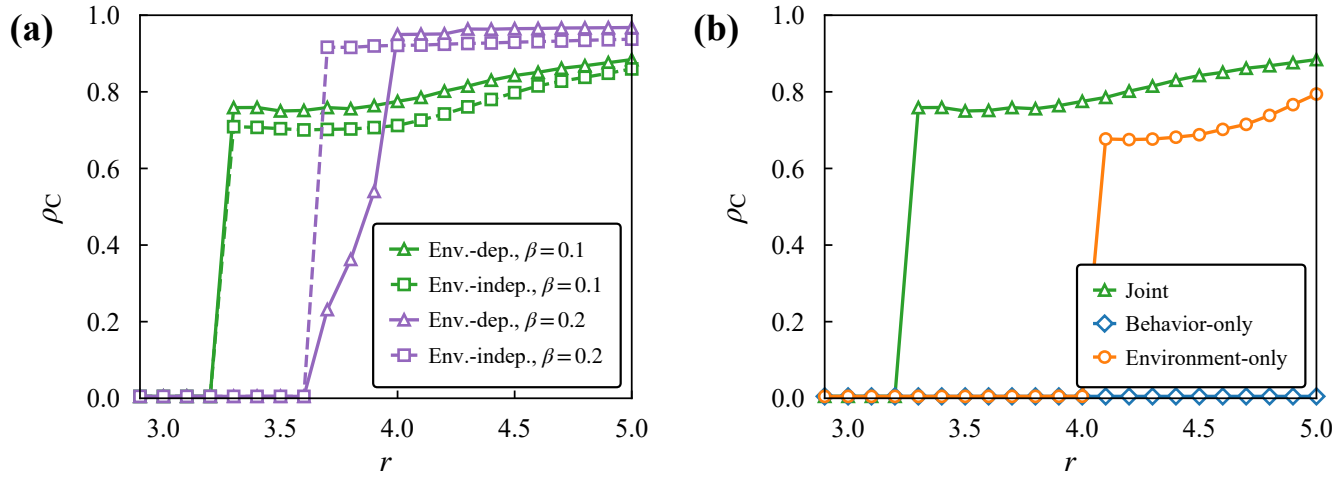


FIG. 6. Neither pressure rule is better in all cases, and the joint state representation supports cooperation over the widest range of synergy factors. (a) Cooperation level ρ_C as a function of r under environment-dependent pressure (solid lines) and environment-independent pressure (dashed lines) for $\beta = 0.1$ and $\beta = 0.2$. Both rules use the joint state representation. (b) ρ_C as a function of r for the joint, behavior-only, and environment-only state representations under environment-dependent pressure at $\beta = 0.1$.

Both rules use the same four states, so that only the reward differs. At $\beta = 0.1$, the two rules give the same onset of cooperation, and the environment-dependent rule yields a slightly higher cooperation level. At $\beta = 0.2$, cooperation emerges at a lower r under the environment-independent rule, whereas the environment-dependent rule yields a slightly higher cooperation level at larger r. The environment-dependent rule is therefore not better in all cases.

This result can be explained by the reward difference in a single interaction. Suppose that k of the $g-1$ co-players of a focal individual cooperate. Under the environment-independent rule, the effective reward of cooperation minus that of defection is

$$\Delta_k^{\mathrm{ind}} = \frac{r}{g} - 1 + \beta\,(2k - g + 1), \tag{8}$$

because changing the action removes the pressure from one side and adds the pressure from the other side. For $g = 5$ and $T = 3$, Eqs. (A.1) and (8) are identical for $k = 0$, 2, and 4. They differ for $k = 1$ and $k = 3$, where the pressure term is $\mp 3\beta$ under the environment-dependent rule and $\mp 2\beta$ under the environment-independent rule. The environment-dependent rule therefore gives a stronger incentive to follow the local majority. This raises cooperation where cooperative groups are common and lowers it where they are rare, which is consistent with Fig. 6(a).

Figure 6(b) compares the three state representations at $\beta = 0.1$. With behavior-only states, cooperation remains close to zero throughout the examined range of r. With environment-only states, cooperation appears only at large r. The joint rep-

resentation supports cooperation at lower r and yields higher cooperation levels. The behavior-only representation retains the two states C and D but omits the group environment, whereas the environment-only representation retains G and B but omits the action of the individual. Under the present pressure rule, the reward of an action depends on the environment, so individuals without environmental information cannot learn different actions for different environments. Retaining both types of information allows individuals to learn action values separately for each combination of action and environment. These results show that distinguishing both the individual action and the group environment during learning helps sustain cooperation over a broader range of synergy factors.

IV. CONCLUSION

Peer pressure is an important form of social influence on cooperation [34, 36–38]. In previous models, the pressure rule does not change with the local behavioral environment, and how such pressure interacts with individual learning [50, 51] has remained unclear. In this work, we have proposed an environment-dependent peer pressure mechanism in a spatial public goods game with Q-learning. Defectors receive pressure in groups with enough cooperators, and cooperators receive pressure in the other groups. The state of each individual includes both its own action and the environment of its current group, so that individuals can learn different actions for different environments.

Our results reveal several findings. First, the effect of peer pressure depends on its intensity and on the synergy factor. Weak pressure allows cooperation to emerge at lower synergy factors, whereas strong pressure yields higher cooperation levels at higher synergy factors. For a given synergy factor, there is an optimal pressure intensity, beyond which cooperation collapses abruptly. Higher synergy factors allow cooperation to persist under stronger pressure. Second, the spatial patterns and the learning dynamics explain these results. Under moderate pressure, cooperative clusters survive the initial decline, and individuals learn to cooperate in cooperative environments. Under strong pressure, cooperative environments are lost at the early stage, which restricts the opportunities for learning a preference for cooperation. Third, the environmental threshold changes the response of cooperation to pressure. Intermediate thresholds favor cooperation under weak and moderate pressure, whereas lower thresholds are more favorable under strong pressure. Finally, the environment-dependent rule is not always better than the environment-independent rule, and the joint state representation supports cooperation over a wider range of synergy factors than the behavior-only or the environment-only representation.

These findings show that the effect of peer pressure on cooperation is determined jointly by the pressure rule, the local environment, and the learning process of individuals. The present results are obtained on a square lattice with fixed learning parameters and random initial conditions. Future studies could consider more complex network structures [17, 54], alternative state-transition rules [69], and separate thresholds for the pressure rule and for the state definition.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

[Author contributions to be added.]

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Appendix: Reward difference in a single interaction

Here we compare the effective rewards of cooperation and defection in a single group interaction. Consider a focal individual and suppose that k of its $g-1$ co-players cooperate, where $k = 0, 1, \ldots, g-1$. From Eqs. (1)–(3), the effective reward of cooperation minus that of defection is

$$\Delta_k = \begin{cases} \dfrac{r}{g} - 1 + \beta k, & k \geq T, \\ \dfrac{r}{g} - 1, & k = T-1, \\ \dfrac{r}{g} - 1 - \beta\,(g-1-k), & k \leq T-2. \end{cases} \tag{A.1}$$

For $k \geq T$, the group is a cooperative environment whichever action the focal individual takes, and a defecting focal individual receives pressure from k cooperators. For $k = T-1$, the group is a cooperative environment if the focal individual cooperates and a defective environment if it defects. The focal individual receives no pressure in either case, and only the cost of cooperation, $1 - r/g$, remains. For $k \leq T-2$, the group is a defective environment whichever action is taken, and a cooperating focal individual receives pressure from $g-1-k$ defectors.

Equation (A.1) shows that peer pressure raises the reward of cooperation relative to defection only when enough co-players

already cooperate, and lowers it otherwise. For $T \leq g-1$, cooperation gives the higher reward when all co-players cooperate if $\Delta_{g-1} > 0$, that is, if

$$\beta > \frac{g-r}{g(g-1)}. \tag{A.2}$$

In contrast, defection always gives the higher reward when all co-players defect, as long as $r < g$. Note that Eq. (A.1) describes the reward of a single interaction only. Individuals using Q-learning also take future rewards into account, and cooperation can be maintained at $\beta = 0$ (Fig. 2), where $\Delta_k < 0$ for all k. We therefore use Eq. (A.1) to interpret the simulation results qualitatively, and not to predict the transition points.

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